Disclaimer

This document is an exam summary and follows the given material of the lecture *Statistical Learning Theory*. Its contribution is a short summary that contains the most important concepts, formulas and algorithms. Due to curriculum content updates, some content may not be relevant to future versions of the course.

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Statistical Learning Theory

Timo Streule, tstreule@ethz.ch (06.02.2022)

1. Basics

- General p-norm: $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$
- Taylor: $f(x) \approx f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + ..$
  - $f(\mathbf{x}) \approx f(\mathbf{a}) + \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}}\big|_{\mathbf{a}} - \frac{1}{2}(\mathbf{x}-\mathbf{a})^\top \frac{\partial^2 f(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^\top}\big|_{\mathbf{a}} (\mathbf{x}-\mathbf{a})$
  - Power series of exp.: $\exp(x) := \sum_{k=0}^{\infty} \frac{x^k}{k!}$
- Entropy: $H(X) \equiv H(p_X) = \mathbb{E}_X[-\log \mathbb{P}(X=x)]$
  - $H(X \mid Y) = \sum_y \mathbb{P}(Y=y) H(X \mid Y=y) \le H(X)$
  - $H(X,Y) = H(X) + H(Y \mid X)$
  - $H(X \mid g(X)) \ge 0$ $\quad$ $H(g(X) \mid X) = 0$
  - $H(5X) \begin{cases} = H(X) & \text{discrete} \\ > H(X) & \text{continuous} \end{cases}$
- MI: $I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y,Z)$ (symmetric)
  - $I(X;Y) = D_{\mathrm{KL}}(p(x,y) \,\|\, p(x)p(y)) \ge 0$
  - $I(X_1,\dots,X_n;Z) = \sum_{i=1}^n I(X_i;Z \mid X_1,\dots,X_{i-1})$
    Markov chain: $I(X_1;X_2,X_3,\dots) = I(X_1;X_2)$
  - $I(X,Y;Z) = I(X;Z) + I(Y;Z \mid X)$
- KL-divergence: $D_{\mathrm{KL}}(p\,\|\,q) = \sum_x p(x) \log\left(\frac{p(x)}{q(x)}\right) \ge 0$
- Cauchy-Schwarz: $|\mathbb{E}[X,Y]|^2 \le \mathbb{E}[X^2]\,\mathbb{E}[Y^2]$
- $1 - z \le \exp(-z)$
- Jensen, $f(X)$ convex: $f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$

1.1. Probability / Statistics

- Gaussian: $\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$
  $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\mu)\right)$
  - $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), Y = A + BX \implies Y \sim \mathcal{N}(A + B\boldsymbol{\mu}, B\boldsymbol{\Sigma}B^\top)$
- Binomial: $f(k,n;p) = \mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$
- $\mathbb{V}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$
  $\mathbb{V}[X+Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2\,\mathbb{C}\mathrm{ov}(X,Y)$
- $\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$
  $= \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$
  $\mathbb{C}\mathrm{ov}(aX,bY) = ab\,\mathbb{C}\mathrm{ov}(X,Y)$

1.2. Calculus

- Partial: $\int uv' \,\mathrm{d}x = uv - \int u'v \,\mathrm{d}x$ $\quad$ $\frac{\partial}{\partial x}\frac{g}{h} = \frac{g'h}{h^2} - \frac{gh'}{h^2}$
- $\frac{\partial}{\partial \mathbf{x}}(\|\mathbf{x}-\mathbf{b}\|_2) = \frac{\mathbf{x}-\mathbf{b}}{\|\mathbf{x}-\mathbf{b}\|_2}$ $\quad$ $\frac{\mathrm{d}}{\mathrm{d}x}|x| = \frac{x}{|x|}$
- $\frac{\partial}{\partial \mathbf{X}} \log|\mathbf{X}| = \mathbf{X}^{-\top}$ $\quad$ $|\mathbf{X}^{-1}| = |\mathbf{X}|^{-1}$
- $\frac{\partial}{\partial \mathbf{x}}(\mathbf{b}^\top \mathbf{x}) = \frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{b}) = \mathbf{b}$
- $\frac{\partial}{\partial \mathbf{x}}(\mathbf{b}^\top \mathbf{A}\mathbf{x}) = \mathbf{A}^\top \mathbf{b}$ $\quad$ $\frac{\partial}{\partial \mathbf{X}}(\mathbf{c}^\top \mathbf{X} \mathbf{b}) = \mathbf{c}\mathbf{b}^\top$
- $\frac{\partial}{\partial \mathbf{X}}(\mathbf{c}^\top \mathbf{X}^\top \mathbf{b}) = \mathbf{b}\mathbf{c}^\top$ $\quad$ $\frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{x}) = 2\mathbf{x}$
- $\frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{A}\mathbf{x}) = (\mathbf{A}^\top + \mathbf{A})\mathbf{x} \overset{\mathbf{A}\text{ sym.}}{=} 2\mathbf{A}\mathbf{x}$
- $\frac{\partial}{\partial \mathbf{X}} Tr(\mathbf{X}^\top \mathbf{A}) = \mathbf{A}$ $\quad$ • Trace trick: $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \dots$
  $\dots \overset{\text{inn. prod.}}{=} Tr(\mathbf{x}^\top \mathbf{A}\mathbf{x}) \overset{\text{cycl. permut.}}{=} Tr(\mathbf{x}\mathbf{x}^\top \mathbf{A}) = Tr(\mathbf{A}\mathbf{x}\mathbf{x}^\top)$
- $\sigma(x) = \frac{1}{1+\exp(-x)} \implies \nabla\sigma(x) = \sigma(x)(1-\sigma(x))$
- $\tanh(x) = \frac{2\sinh(x)}{2\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ $\quad$ •
  $\nabla \tanh(x) = 1 - \tanh^2(x)$

2. Empirical Risk Minimisation (ERM)

Cost: $R(c,X,Y) = \sum_{i \le N} \|y_i - c^\top \mathbf{x}_i\|^2$ (regr.)
or $R(c,X,Y) = \sum_{i \le N} \max(0, -y_i c^\top \mathbf{x})$ (class.)
or $R(c,\theta,X) = \sum_{i \le N} \|\mathbf{x}_i - \theta_{c(i)}\|^2$ (clust.)
Goal: $\arg\min_c \mathbb{E}_{\mathcal{X}}[R(c,\mathcal{X}] \approx \arg\min_c \frac{1}{N} R(c,X)$

2.1. Bayesianism / Frequentism

Bayesianism: Define prior $P(\theta)$, define likelihood $P(X \mid \theta)$, compute posterior $P(\theta \mid x_{1\dots n})$.
Bayes: $P(\theta \mid X) = \frac{P(X \mid \theta) P(\theta)}{P(X)}$, $P(X) = \sum_\theta P(X \mid \theta_i) P(\theta_i)$
Frequentism: Define param. model $P(Y \mid X, \theta)$, compute likelihood of data $P((X,Y) \mid \theta)$ and compute $\hat{\theta}_{\mathrm{MLE}}$ via $\arg\max_\theta$ of likelihood.

2.2. Linear Regression

model: $\hat{\mathrm{y}} = \mathbf{X}\beta$

Ridge: $\epsilon_{\mathrm{RSS}}(\beta,\lambda) = (\mathbf{y} - \mathbf{X}^\top\beta)^\top(\mathbf{y} - \mathbf{X}^\top\beta) + \lambda\beta^\top\beta$
$\hat{\beta} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbb{I})^{-1}\mathbf{X}^\top\mathbf{y}$, prior: $\beta \sim \mathcal{N}(0, \frac{\sigma^2}{\lambda}\mathbb{I})$
Lasso: $\hat{\beta} = \arg\min_\beta \sum_{i \le n}(y_i - x_i^\top\beta)^2 + \lambda\|\beta\|_2$
(no closed form), prior: $p(\beta_i) = \frac{\lambda}{4\sigma^2} \exp(-|\beta_i| \frac{\lambda}{2\sigma^2})$

3. Maximum Entropy Inference

Sample $c \sim p(\cdot \mid X)$ s.t. $H[p(\cdot \mid x)]$ is maximal, $\mathbb{E}_{C|X}[R(C,X)] = \mu$ and $\sum_c p(c \mid X) = 1$.
$\implies$ **Gibbs dist.:** $p(c \mid X) = \frac{1}{Z(X)} \exp(-\beta R(c,X))$
Free energy: $F(X) := -\frac{1}{\beta}\log Z(X)$
$\iff$ $p(c \mid X) = \exp(-\beta[R(c,X) - F(X)])$
$\implies$ **entropy:** $H[c \mid X] = \beta \underbrace{\mathbb{E}_{C|X}[R(C,X)]}_{=\mu} - \beta F(X)$
ME: $\max H[c \mid X] \iff \max Z(X) \iff \min F(X)$

- Exp. generalisation costs: $\mathbb{E}_{X''}\mathbb{E}_{X'}\mathbb{E}_{C|X'}[R(c,X'')]$ ($\mathbb{E}_C$)
- Min. out-of-sample descr. length per deg. of freedom
  $\min_{p(\cdot|\cdot)} \mathbb{E}_{X',X''}\mathbb{E}_{C|X'}\left[-\log\frac{p(c|X'')}{p(c)}\right] p(c) = \mathbb{E}_X[p(c \mid X)]$
  $\overset{\text{Jensen}}{\ge} \min_{p(\cdot|\cdot)} \mathbb{E}_{X',X''}\left[-\log \mathbb{E}_{C|X'}[p(c \mid X'')]\right] - H[c]$
  $= \max_{p(\cdot|\cdot)} \mathbb{E}_{X',X''}[e^{H[c]} \cdot \kappa(X',X'')]$

PA: $T^* = \arg\max_T \kappa(X',X'')$
- PA-kernel: $\kappa(X',X'') := \sum_c p(c \mid X') p(c \mid X'')$
- combined: $p(c \mid X',X'') \propto p(c \mid X') p(c \mid X'')$

4. Methods for intractable Gibbs distr.

4.1. Sampling and SA

Well behaving **Markov Chains** are
- **irreducible:** can go from/to any state, and
- **aperiodic:** doesn't go "back&forth" forever.

$\implies$ **Stationary dist.** $p(c') = \sum_c \pi(c \mid c') p(c)$
$\iff$ **det. balance** $\pi(c' \mid c) p(c) = \pi(c \mid c') p(c')$

Metropolis-Hastings: Assume $p(c) \propto f(c)$.

$$\pi(c' \mid c) := \begin{cases} q(c' \mid c)\, A(c,c') & c \ne c' \\ 1 - \sum_{c' \ne c} q(c' \mid c)\, A(c,c') & \text{otw.} \end{cases}$$

where $q(c' \mid c)$: prob. to propose the move $c \to c'$, and $A(c,c') := \min\left\{1, \frac{q(c|c')\, f(c')/\mathcal{Z}}{q(c'|c)\, f(c)/\mathcal{Z}}\right\}$ prob. accept move

Metropolis Algorithm: Assume $p(c) \propto f(c)$ and

$q(c' \mid c) = q(c \mid c')$, i.e. symmetric.

1. Define symmetric $\{q(\cdot \mid c)\}_{c \in \mathcal{C}}$ s.t. graph G_q is connected and every vertex in G_q has edge to itself.
2. $c_0 \leftarrow \$$ Then, for $t = 1, 2, \ldots$, do:
   - $\tilde{c} \leftarrow q(\cdot \mid c_{t-1})$ // sample
   - $b \leftarrow \text{Bern}\left(\min\left\{1, \mathrm{e}^{-\frac{1}{T}[R(\tilde{c},X) - R(c_{t-1},X)]}\right\}\right)$
   - If $b = 1$ then $c_t \leftarrow \tilde{c}$ else $c_t \leftarrow c_{t-1}$.

$\pi(c' \mid c) = \{\cdots$ ← c.f. scr. (2.7)

Simulated annealing: Gradually decrease temp. T to escape bad local minima. → MH-sampling from Gibbs (DA does not sample!).

4.2. Laplace's Method (Least angle clust.)

1. *Square the cost:* $\mathrm{e}^{-\frac{1}{T}R(c,X)} = \text{const} \cdot \mathrm{e}^{g(c)^\top g(c)}$
2. *Complete the square:* $\int \mathrm{e}^{-\frac{1}{T}(y-g(c))^2}\,\mathrm{d}y = (\pi T)^{d/2}$
   $\Rightarrow \mathrm{e}^{g(c)^\top g(c)} = (\pi T)^{-d/2} \int \exp^{-y^\top y + 2y^\top g(c)}\,\mathrm{d}y$
3. *Rewrite normalisation constant:*
   $Z = \sum_c \mathrm{e}^{-\frac{1}{T}R(c,X)} = \ldots = \text{const} \int \mathrm{e}^{-\frac{1}{T}f(y)}\,\mathrm{d}y$
4. *Apply Laplace's method:*
   If f has unique min. y_0 and Hessian $H := \frac{\partial^2 f}{\partial y^2}\big|_{y_0}$
   $\int \mathrm{e}^{-\frac{1}{T}f(y)}\,\mathrm{d}y \overset{(T\to 0)}{\approx} \mathrm{e}^{-\frac{1}{T}f(y_0)} \left|\frac{H}{2\pi T}\right|^{-1/2}$

4.3. Mean-field Approximation

Idea: Approximate p_β (Gibbs) with a "simple", factorisable distribution $p = p_1 \cdots p_N$.

Approach: Minimise $D_{\mathrm{KL}}(p \parallel p_\beta)$
$\iff$ Minimise **Gibbs free energy:**
$G(p) = \frac{1}{\beta} D_{\mathrm{KL}}(p \parallel p_\beta) + F(\beta) = \mathbb{E}_{c\sim p}[R(c)] - \frac{1}{\beta} H[p]$

Note: $H[p] = \sum_{i=1}^N H[p_i]$ *and* $F(\beta) \le G(p)$

Ising model: $R(c \mid J) = -\frac{1}{2}\sum_{i,j} J_{ij} c_i c_j - \sum_i h_i c_i$
where J_{ij}: interaction between particles,
h_i: noisy image, σ_i: denoised image

Problem: $\frac{\partial G(p)}{\partial p_{i\ell}} = 0$ s.t. $\sum_{\ell'} p_{i\ell'} = 1 \; \forall i$

Solution: with the *mean field* $h_i = [\cdots h_{i\ell} \cdots]^\top$

$h_{i\ell} := \frac{\partial \mathbb{E}[R(c)]}{\partial p_{i\ell}} = \mathbb{E}_{c\sim p_{|i\to\ell}}[R(c)]$ ← object i chooses class ℓ

$p_{i\ell} = \mathrm{e}^{-\beta h_{i\ell}} / Z_i$

EM-like Algo: Iteratively 1. Pick random i 2. $h_i^{\text{new}} \leftarrow p_j^{\text{old}}$ 3. $p_i^{\text{new}} \leftarrow h_i^{\text{new}}$ until converged.

4.3.1. Smooth *k*-means scr.20 (p. 39)

$R(c \mid X) = \sum_i \|x_i - y_{c_i}\|^2 + \frac{\lambda}{2}\sum_i \sum_{j\in N(i)} \mathbb{I}_{\{c_i \neq c_j\}}$
where the second term measures #violations of these neighbourhood constraints.

$\implies h_{i\ell} = \|x_i - y_\ell\|^2 + \lambda \sum_{j\in N(i)} p_{j\ell} + \text{const}_i$

5. Deterministic Annealing (Z is tractable)

Lemma: func's × domain → domain × co-dom.
$\mathcal{O}(K^N) \to \sum_c \prod_i \epsilon_{i,c(i)} = \prod_i \sum_k \epsilon_{ik} \leftarrow \mathcal{O}(NK)$
$p(c \mid \theta, X) = \prod_{i\le N} p_i(c(i) \mid \theta, X)$
where $p_i(k \mid \theta, X) \propto \exp(-\frac{1}{T}\|x_i - \theta_k\|^2)$

Max. entr. $\implies \frac{\partial \log Z}{\partial \theta_k} = 0 \implies \theta_k^* = \frac{\sum_i p_i(k|\theta^*,X)\cdot x_i}{\sum_i p_i(k|\theta^*,X)}$

do
  E-step: $p_i(k|\theta^{\text{old}}, X) = \frac{\exp(-\frac{1}{T}\|x_i - \theta_k\|^2)}{\sum_{j\le K}\exp(-\frac{1}{T}\|x_i - \theta_j\|^2)}$
  M-step: $\theta_k \leftarrow \ldots$
  $\theta^{\text{old}} \leftarrow \theta$
until convergence of θ
$\theta_k \leftarrow \theta_k + \epsilon$ (noise s.t. centroids can separate)

Phase transitions: For $T \to \infty$: $\theta_k^* = \bar{\mathbf{X}} \;\; \forall k \le K$
Once $T = 2\lambda_{\max}$, more centroids appear, where $\lambda_{\max}$ = max. eigenvalue of $\frac{1}{N}\mathbf{X}^\top\mathbf{X}$. ($\mathbf{x}_i$'s row-wise)

6. Histogram Clustering

Least Angle Clust. (LAC): [Idea]
Similarity $S(\mathbf{x}_i, \mathbf{x}_j) = w_{ij}\cos(\phi_{ij}) = w_{ij}\mathbf{e}_i \cdot \mathbf{e}_j$ with unit vectors $\mathbf{e}_i := \mathbf{x}_i/\|\mathbf{x}_i\|$, e.g. choice $w_{ij} = \|\mathbf{x}_i\| \cdot \|x_j\|$.

Dyadic data: $\mathcal{Z} = \{(x_{i(r)}, y_{j(r)}); 1 \le r \le \ell\}$
- prototype / "centroid": $q(y_j \mid \alpha)$
- empirical dist.: $\hat{p}(y_j \mid x_i) = \frac{\hat{p}(x_i,y_j)}{\hat{p}(x_i)}$ ←scr. (5.10) ←scr. (5.11)

Likelihood: $P(\mathcal{Z} \mid c, q) = \prod_{r\le\ell} p(x_{i(r)}, y_{j(r)} \mid c, q)$
$= \overset{\text{scr. (5.12)}}{\cdots} = \prod_i \prod_j [q(y_j|c(i)) \cdot p(c(i))]^{\ell \hat{p}(x_i,y_i)}$
Assume $p(\alpha) = 1/k$ *and* $\hat{p}(x_i) = 1/n$
$\Rightarrow$ **Cost:** $R^{\text{hc}}(c,q,\mathcal{Z}) = \frac{\ell}{n}\sum_{i\le n} D_{\mathrm{KL}}[\hat{p}(\cdot \mid x_i) \parallel q(\cdot \mid c(i))]$
Solving the **Gibbs dist.** $p(c \mid q, \hat{p}) = \prod_{i\le n} P_{i,c(i)}$
via Lagrange yields $q^*(y_j \mid \alpha) = \frac{\sum_{i\le n} P_{i\alpha}\cdot \hat{p}(y_j|x_i)}{\sum_{i\le n} P_{i\alpha}}$
Lemma 2 ch.3 p.36

6.1. Information Bottleneck Method

Find efficient code $X \mapsto \hat{X}$ (codebook vector) and preserve relevant info. about context Y.

Criterion: $R^{\text{IB}}(q(\hat{x} \mid x)) = I(X;\hat{X}) - \beta I(\hat{X};Y)$

Markov chain: $\hat{X} \xrightarrow{q(\hat{x}|x)} X \xrightarrow{p(y|x)} Y$

Generation process: w/ *distortion* $d(x,\hat{x}) = D_{\mathrm{KL}}[\cdot]$

$$\begin{cases} q_t(\hat{x}|x) & \propto q_t(\hat{x}) \cdot \exp(-\beta\, D_{\mathrm{KL}}[p(y|x) \parallel p_t(y|\hat{x})]) \\ q_{t+1}(\hat{x}) & = \sum_x p(x) \cdot q_t(\hat{x} \mid x) \\ p_{t+1}(y|\hat{x}) & = \sum_x p(y \mid x) \cdot p(x) \cdot q_t(\hat{x} \mid x) \,/\, q_t(\hat{x}) \end{cases}$$

6.2. Parametric Distributional Clustering

Idea: Use a mixture of Gaussian prototypes, i.e.
$p(y_j \mid \nu) \equiv p(b \mid \nu) = \sum_{\alpha\le s} p(\alpha|\nu)\, G_\alpha(b)$.
$x_i \xrightarrow{c(i)=\nu} \nu \xrightarrow{p(b|\nu)} \hat{p}(b \mid i)$

Note: Feature values y_j ("bins" b) only depend on cluster index ν and not explicitly on the site x_i!

Notation: $x_i \leftarrow i$, $y_j \leftarrow b$ (*bins*), $\nu \leftarrow$ clusters

Likelihood: (both equivalent if $p(i) = \frac{1}{n}$)
$P(X \mid c, \theta) = \prod_{i\le n} p(c(i)) \prod_{b\le m} [p(b \mid c(i))]^{\ell \hat{p}(i,b)}$,
$P(X, M \mid \theta) = \prod_{i\le n} \prod_{\nu\le k} [p(\nu) \cdot \prod_{b\le m} p(b \mid \nu)^{n_{ib}}]^{M_{i\nu}}$
where n_{ib} : #occur. an observ. at site i is inside I_b
$M_{i\nu} = p(\nu \mid i) \in \{0,1\}$ clust. membersh. assign.

Cost (IB): $R^{\text{PDC}}(c, p_{\cdot|c}) = -\log P(X, M\theta) = \ldots \ldots = -\sum_{i\le n}\left[\log p_{c(i)} + \frac{\ell}{n}\sum_{b\le m}\hat{p}(b\mid i)\log p(b\mid c(i))\right]$

E-step: $h_{i\nu} = -\log p_\nu - \sum_b \frac{\ell}{n}\hat{p}(b\mid i)\log p(b\mid \nu)$
$q_{i\nu} = \mathbb{E}[\mathbb{I}_{\{c(i)=\nu\}}] \propto \exp(-h_{i\nu}/T)$

M-step: $p_\nu = \frac{1}{n}\sum_{i\le n} q_{i\nu}$
No closed form sol. for $p(\alpha\mid\nu)$. Thus, iteratively optimize pairs s.t. $\sum_\alpha p(\alpha\mid\nu) = 1$.

7. Graph-based Clustering

Non-metric relations: might assume negative values or violate the triangular inequality.

Setting: objects $\mathbf{o}_i, \mathbf{o}_j \in \mathcal{O}$; relations with weights $\mathcal{D} := \{D_{ij}\}$ on the edges (i,j).

- Cluster α: $\mathcal{G}_\alpha \equiv \{\mathbf{o}\in\mathcal{O} : c(\mathbf{o}) = \alpha\}$
- Inter-cluster edges: $\mathcal{E}_{\alpha\beta} = \{(i,j)\in\mathcal{E} : \mathbf{o}_i\in\mathcal{G}_\alpha \wedge \mathbf{o}_j \in \mathcal{G}_\beta\}$
- $\text{cut}(A,B) = \sum_{i\in A, j\in B} W_{ij}$ → weight matrix W
- $\text{assoc}(A,\mathcal{V}) = \sum_{i\in A, j\in\mathcal{V}} W_{ij}$ → total connection strength from nodes in A to all nodes in the graph

Correlation clustering:

Minimise the sum of *pairwise* intracluster distances.

$$R^{\text{cc}}(c;\mathcal{D}) = -\sum_{\nu\le k}\sum_{(i,j)\in\mathcal{E}_{\nu\nu}} S_{ij} + \sum_{\nu\le k}\sum_{\substack{\mu\le k\\ \mu\neq\nu}}\sum_{(i,j)\in\mathcal{E}_{\nu\mu}} S_{ij}$$
$$= -2\sum_{\nu\le k}\sum_{(i,j)\in\mathcal{E}_{\nu\nu}} S_{ij} + \cancel{\sum_{(i,j)} S_{ij}}$$
↪ intra-cluster ↪ const

up to thresh. u:
$$\overset{*}{=} -\tfrac{1}{2}\sum_{\nu\le k}\sum_{(i,j)\in\mathcal{E}_{\nu\nu}} (|S_{ij}-u| + S_{ij} - u) + \tfrac{1}{2}\sum_{\nu\le k}\sum_{\substack{\mu\le k\\ \mu\neq\nu}}\sum_{(i,j)\in\mathcal{E}_{\nu\mu}} (|S_{ij}+u| - S_{ij} - u)$$

$*$: altern. def. where $\frac{1}{2}(|X| \pm X) = \max\{0, \pm X\}$

Graph partitioning: $D_{ij}\in\mathbb{R}$

$$R^{\text{gp}}(c;\mathcal{D}) = \text{const} - \sum_{\nu\le k}\text{cut}(\mathcal{G}_\nu(\mathcal{D}), \mathcal{V}\ \mathcal{G}_\nu(\mathcal{D})) = \text{const} + \sum_{\nu\le k}\text{cut}(\mathcal{G}_\nu(\mathcal{S}), \mathcal{V}\ \mathcal{G}_\nu(\mathcal{S}))$$

Bias in $R(c;D)$: Cost should scale prop. to #objects, i.e. $R(c;D) = \mathcal{O}(n)$. $*$: use $D_{ij} = D(1-\delta_{ij})$

Tipp: $\frac{\text{cut}(\mathcal{G}_\alpha, \mathcal{V}\setminus\mathcal{G}_\alpha)}{\text{assoc}(\mathcal{G}_\alpha,\mathcal{V})} \overset{*}{=} \frac{n\cdot p_\alpha\cdot n(1-p_\alpha)\cdot D}{n\cdot p_\alpha\cdot n\cdot D} = 1 - p_\alpha$

7.1. Pairwise Clustering

Cost: $R^{\text{pc}}(c;\mathcal{D}) = \sum_\alpha \sum_{(i,j)\in\mathcal{E}_{\alpha\alpha}} \frac{D_{ij}}{|\mathcal{G}_\alpha|} = \sum_\alpha \sum_{(i,j)\in\mathcal{E}_{\alpha\alpha}} |\mathcal{G}_\alpha| \frac{D_{ij}}{|\mathcal{E}_{\alpha\alpha}|}$

Equivariance to k-means: (if $D_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|^2$)

$$\sum_{i\le n}\|\mathbf{x}_i - \mathbf{y}_{c(i)}\|^2 = \sum_{i\le n}\sum_{j\le n}\sum_{\alpha\le k}\frac{\mathbb{I}_{\{c(i)=\alpha\}}\mathbb{I}_{\{c(j)=\alpha\}}}{|\mathcal{G}_\alpha|} D_{ij}$$

Invariance properties:

- Symmetrisation: $R^{\text{pc}}(c;\mathcal{D}^{\text{s}}) \equiv R^{\text{pc}}(c;\mathcal{D})$
- Off-diagonal shift: $R^{\text{pc}}(c;\tilde{\mathcal{D}}) = R^{\text{pc}}(c;\mathcal{D}) - \lambda_{\min}\cdot n$

Theorem: If S^{c} is p.s.d., then D derives from squared Eucl. space. $\implies$ Make S **p.s.d.**: $\tilde{S} := S - \lambda_{\min}\mathbb{I}$

Constant Shift Embedding:

1. **Symmetrise** $D \to D^{\text{s}}$: $D^{\text{s}}_{ij} := \frac{1}{2}(D_{ij} + D_{ji})$
2. **Centralise** D, then S: $X^{\text{c}} := QX^{\text{s}}Q^\top$
   $Q = \mathbb{I} - \frac{1}{n}\mathbf{e}_n\mathbf{e}_n^\top$ $\quad S^{\text{c}} = -\frac{1}{2}D^{\text{c}}$
   $X^{\text{c}}_{ij} = X_{ij} - \frac{1}{n}\sum_k X_{ik} - \frac{1}{n}\sum_k X_{kj} + \frac{1}{n^2}\sum_{k,\ell} X_{k\ell}$
   $\implies$ sum over column/rows = 0
3. **(Off-)Diagonal shift**: Find $\lambda_{\min}$ of S^{c}
   $\tilde{S} := S^{\text{c}} - \lambda_{\min}\mathbb{I}$ $\quad \tilde{D} := D - \lambda_{\min}(\mathbf{1} - \mathbb{I})$
   $\tilde{D}_{ij} = \tilde{S}_{ii} + \tilde{S}_{jj} - 2\tilde{S}_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|^2$

Reconstruction:

1. EVD: $\tilde{S} = \mathbf{V}\Lambda\mathbf{V}^\top$ via $(\tilde{S} - \lambda\mathbb{I})\mathbf{v} \overset{!}{=} 0$ $(|\mathbf{v}| = 1)$
   where $\Lambda = \text{diag}(\lambda_1 \ldots \lambda_n)$ and $\mathbf{V} = [\mathbf{v}_1 \ldots \mathbf{v}_n]$
2. Find p s.t. $\lambda_1 \ge \ldots \lambda_p > \lambda_{p+1} = \ldots = \lambda_n = 0$
3. $\implies$ $\mathbf{X}_p = \mathbf{V}_p(\Lambda_p)^{1/2}$ (each row is a vector)
4. $\implies$ $\mathbf{X}_t = \mathbf{V}_t(\Lambda_t)^{1/2}$ (approx. & denoising)

Cluster membership of new data:

Note: S^{new} is def. by $D^{\text{new}}_{ij} = S^{\text{new}}_{ii} + \tilde{S}_{jj} - 2S^{\text{new}}_{ij}$

1. $(S^{\text{new}})^{\text{c}} = -\frac{1}{2}[D^{\text{new}}(\mathbb{I}_n - \frac{1}{n}\mathbf{e}_n\mathbf{e}_n^\top) - \frac{1}{n}\mathbf{e}_m\mathbf{e}_n^\top + \tilde{D}(\mathbb{I}_n - \frac{1}{n}\mathbf{e}_n\mathbf{e}_n^\top)]$
2. Project: $X^{\text{new}}_p = (S^{\text{new}})^{\text{c}}\mathbf{V}_p(\Lambda_p)^{-1/2}$
3. Assign: $\hat{c}_i = \arg\min_c \left\|(x^{\text{new}}_p)_i - y_{c(i)}\right\|$

8. Model Selection for Clustering

What is the appropriate #clusters k for my data?

General approach: Measure quality (neg. log-likelihood) for different k → **elbow**.

8.1. Complexity-based Model Selection

Strategy: add a complexity term to neg. log-likelihood

Attention: MDL/BIC rely on likelihood optimisation → not generally applicable

Ocam's razor: Choose the model that provides the shortest description of the data.

8.1.1. Min. Description Length (MDL)

Minimise **descr. length**: $-\log p(\mathbf{X}\mid\theta) - \log p(\theta)$

Approx.: $\hat{k} \in \arg\min_k -\log p(\mathbf{X}\mid\hat{\theta}) + \frac{k'}{2}\log n$

8.1.2. Bayesian Information Crit. (BIC)

Parametrise likelihood $p(\mathbf{X}\mid M)$ by θ:

$p(\mathbf{X}\mid M) = \int_{\Theta_M} \exp(\log p(\mathbf{X}\mid M,\theta))\cdot p(\theta\mid M)\,\mathrm{d}\theta$

Assume flat prior $p(\theta|M) \approx \text{const}$ and expand log-likelihood by ML estimator $\hat{\theta}$:

$\bar{\ell}(\theta) = \frac{\ell(\theta)}{n} = \frac{1}{n}\log p(\mathbf{X}|M,\theta) \overset{\text{i.i.d.}}{=} \frac{1}{n}\sum_i \ell(\theta, X_i) \overset{\text{Taylor}}{\approx} \ldots$

$\implies p(\mathbf{X}\mid M) = \text{const}_2\cdot\exp(\ell(\hat{\theta}) - \frac{k'}{2}\log n)$

where k' : dimension of (trainable) parameters

9. Model Validation

9.1. Stability-based Validation

Stability: Solutions on two data sets drawn from the same source should be similar.

9.2. Information-theoretic Validation

9.2.1. Shannon's Channel Coding Thm.

- **Channel:** $(\mathcal{S}, \{p(\cdot \mid s)\}_{s\in\mathcal{S}})$, $\mathcal{S}$: alphabet
  - ϵ-noisy binary channel: $p(\hat{s} \mid s) = \begin{cases} 1-\epsilon & \text{if } \hat{s}=s \\ \epsilon & \text{if } \hat{s}\neq s \end{cases}$
- **Capacity:** $\mathrm{cap} = \max_p I(S;\hat{S}) \rightsquigarrow p_S(s)$
- **$(\mathbf{M},\mathbf{n})$-code:** is a pair (Enc, Dec) ← scr. p.87
  where M: #messages, n: code-length
  - **Rate:** $r = \frac{\log_2 M}{n} \Leftrightarrow M = \lfloor 2^{nr} \rfloor$
  - **Commu. err.:** $p_{\text{err}} := \max_{i\leq M} \mathbb{P}(Dec(\widehat{Enc(i)}) \neq i)$

Goal / **Best code:** $\lim_{n\to\infty} \frac{\log M}{n}$ s.t. $\lim_{n\to\infty} p_{\text{err}} \to 0$

Asymptotic equiparition property (AEP):

- $A_\epsilon^{(n)}$: Typical set of sequences $(s_1,\dots,s_n) \in \mathcal{S}^n$
  $\left|-\frac{1}{n}\log p_{S^n}(\mathbf{s}^n) - H[S]\right| < \epsilon$ ← scr. p.89
- $\mathbb{P}\left((\mathbf{S}^n, \hat{\mathbf{S}}^n) \in A_\epsilon^{(n)}\right) \overset{n\to\infty}{\to} 1$ ← scr. p.90
- $p_{\text{err}} \leq 2^{-n(\mathrm{cap}-3\epsilon-r)} \overset{n\to\infty}{\to} 0$ if $r < \mathrm{cap}$

9.2.2. Algorithm Validation

Assumptions:

- Exponential solution space, i.e. $\log|\mathcal{C}| = \mathcal{O}(n)$
- $\mathcal{A}$'s output is probabilistic, i.e. $p(\cdot \mid X')$

Ideal variant:

Messages: $\mathcal{M} = \{X'_1,\dots,X'_m\}$

Code: $X'_i \xrightarrow{Enc_{\mathcal{A}}} p(\cdot \mid X'_i) \xrightarrow{\mathcal{C}_{\mathcal{A}}} p(\cdot \mid X''_i) \xrightarrow{Dec_{\mathcal{A}}} \hat{X}$

Empirical variant:

Messages: $\mathcal{M} = \{\tau_1,\dots,\tau_m\}$ drawn u.a.r. from $\mathbb{T}$

- Require $\sum_\tau p(c \mid \tau \circ X') \approx \frac{|\mathbb{T}|}{|\mathcal{C}|} \pm \rho$ ← scr. p.95

Code: $\tau_i \xrightarrow{Enc} p(\cdot \mid \tau_i \circ X') \xrightarrow{\mathcal{C}_{\mathcal{A}}} p(\cdot \mid \tau_i \circ X'') \xrightarrow{Dec} \hat{\tau}$

- $Enc_{\mathcal{A}}$: encodes $\tau_i \in \mathcal{M}$ as $p(\cdot \mid \tau_i \circ X')$
- $Dec_{\mathcal{A}}$: selects $\hat{\tau} = \arg\max_\tau \kappa(\tau_i \circ X'', \tau \circ X')$

whereby $\kappa(X'',X') := \sum_c p(c \mid X'')p(c \mid X')$

Asymptotic Equipartition Property (AEP):

AEP fulfilled if $\log\kappa(X',X'') \overset{n\to\infty}{\to} \mathcal{E}$
whereby $\mathcal{E} := \mathbb{E}_{X',X''}[\log\kappa(X',X'')]$

- $A_\epsilon^{(n)}$: set of (ϵ,n)-typical pairs X', X''
  $|\log\kappa(X',X'') - \mathcal{E}| < \epsilon$
- $p_{\text{err}} \leq P_{(n)}$ c.f. scr. (6.19) $\overset{n\to\infty}{\to} 0$ if $\frac{\log m}{\log|\mathcal{C}|} < I$
  where $I := \frac{1}{\log|\mathcal{C}|}\mathbb{E}_{X',X''}[\log(|\mathcal{C}|\kappa(X',X''))]$

9.3. Applications of PA

PA: *quantifies the amount of information that algorithms extract from phenomena.* → quantified by **capacity** (max. # distinguishable messages that can be communicated)

Temperature: $T^* = \arg\max_T \kappa(X',X'')$

Cost functions: Given $R_1(\cdot,\cdot),\dots,R_s(\cdot,\cdot)$
$\max_{\ell\leq s}\kappa_\ell(X',X'') = \max_{\ell\leq s}\frac{1}{Z_{X'}Z_{X''}}\sum_c \mathrm{e}^{-\frac{1}{T}R_\ell(c,X')}\mathrm{e}^{-\frac{1}{T}R_\ell(c,X'')}$

Algorithms: Many MST (min. spanning tree) algo's are **contractive** (→ sequence of candidate sol's).

Approximation Set Coding (ASC):

$$p^{\text{ASC}}(c \mid X') = \begin{cases} 1/|G_\gamma(X')| & \text{if } c \in G_\gamma(X') \\ 0 & \text{otw.} \end{cases}$$

$$G_\gamma(X') := \left\{c \in \mathcal{C} : R(c,X') - \min_{c\in\mathcal{C}} R(c,X') \leq \gamma\right\}$$

1. Run $\mathcal{A}$ to compute $G_t^{\mathcal{A}}(X')$ and $G_t^{\mathcal{A}}(X'')$, for all t

2. $t^* = \arg\max_t \kappa(X',X'') = \arg\max_t \frac{|G_t^{\mathcal{A}}(X') \cap G_t^{\mathcal{A}}(X'')|}{|G_t^{\mathcal{A}}(X')| \cdot |G_t^{\mathcal{A}}(X'')|}$

3. $c^* \xleftarrow{\$\text{ sample}} \mathrm{Unif}(G_{t^*}^{\mathcal{A}}(X') \cap G_{t^*}^{\mathcal{A}}(X''))$

max. 4 A4-pages!